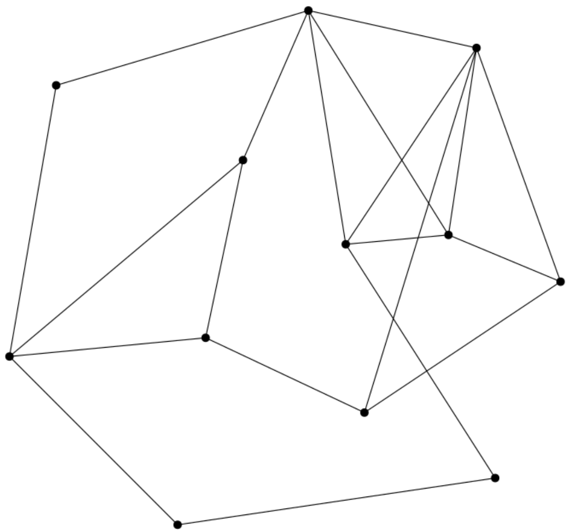
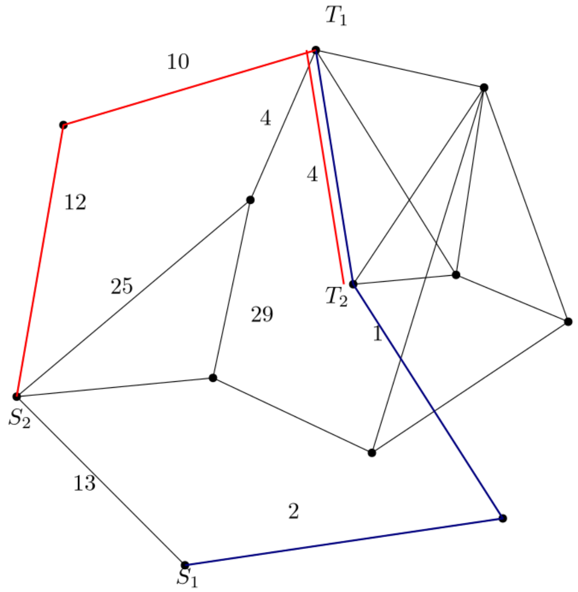
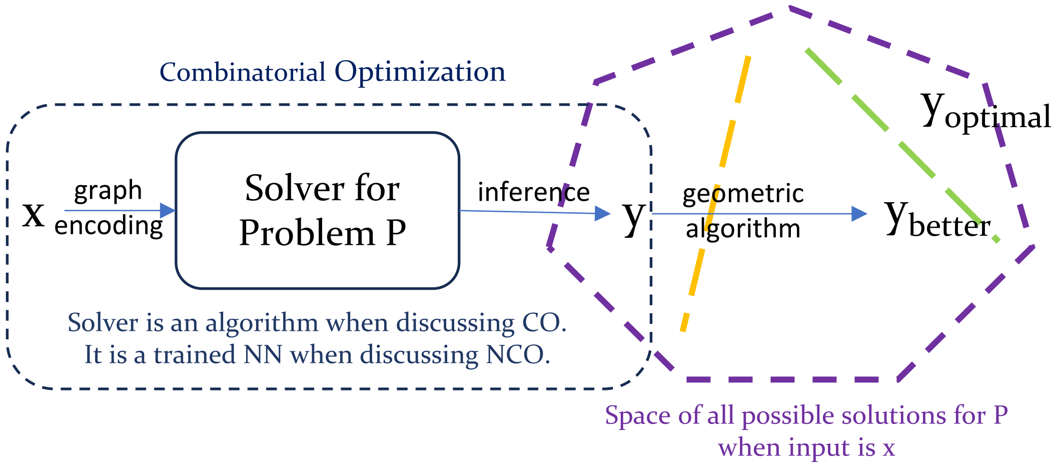
Neural Combinatorial Optimization for Multi Agent Path Finding on Graphs







Graphs

$G = (V, E, w)$

V : vertices

E : edges

w : weight on the edges

Multi Agent Path Finding (MAPF)

$A_1, A_2, \dots, A_k$: agents

$s_1, s_2, \dots, s_k$: starting

$t_1, t_2, \dots, t_k$: target

$\pi_1, \pi_2, \dots, \pi_k$: paths/policy

NP-hard : $O(n^c)$ algorithm unlikely

Neural Combinatorial Optimization (NCO)

CO : mathematical problem whose solution space is a set of discrete choices

NCO : neural networks learns the problem and given an output from the solution space

Geometric NCO : improves solution obtained by exploiting the structure of solution space

Question

Can we train GNNs to obtain better inference (solution) for MAPF on Graphs using Geometric NCO approach?

Base Papers

1. **Geometric Algorithms for Neural Combinatorial Optimization with Constraints (NeurIPS 2025)**
   Nikolaos Karalias, Akbar Rafiey, Yifei Xu, Zhishang Luo, Behrooz Tahmasebi, Connie Jiang, Stefanie Jegelka
2. ***Multi-Agent Pathfinding: Definitions, Variants, and Benchmarks (SoCS 2019)***
   Roni Stern, Nathan R. Sturtevant, Ariel Felner, Sven Koenig, Hang Ma, Thayne T. Walker, Jiaoyang Li, Dor Atzmon, Liron Cohen, T. K. Satish Kumar, Eli Boyarski, Roman Bart´ak
3. ***Graph Attention Networks based Multi-agent Path Finding via Temporal-spatial Information Aggregation (PLOS One 2025)***
   Qingling Zhang, Peng Wang, Cui Ni, Xianchang Liu

Current Scenario

Extended A* , Conflict Based Searches, Generalized RL algorithms

Plan

Specific variation of MAPF (Sept 30th) → Obtain a structure from the solution space (Sept 30th) → Design a good objective function (Sept 30th)

→ Train a GNN → Benchmark (Networks, Heuristic models, Algorithms) (Oct 30th)

Data Set

1. Synthetic or randomly generated graphs.
2. Real road networks (SNAP, movingai, zenodo)